Floor beam B1

Simply supported floor beam · Example building, axis B

logo

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Simply supported beam, uniformly loaded, laterally restrained by the slab. Profile IPE 300, steel S355.

Geometry and loads

Symbol	Expression	Value	Note
L		$6.0\,\mathrm{m}$	span
q_{Ed}		$25\,\mathrm{kN/m}$	design load, ULS 6.10b
q_{ser}		$18\,\mathrm{kN/m}$	characteristic load, SLS

Symbol	Expression	Value	Note
M_{Ed}	$\frac{q_{Ed}\,L^2}{8} = \frac{25\,\mathrm{kN/m}\cdot(6.0\,\mathrm{m})^2}{8}$	$112.5\,\mathrm{kN{\cdot}m}$	
V_{Ed}	$\frac{q_{Ed}\,L}{2} = \frac{25\,\mathrm{kN/m}\cdot 6.0\,\mathrm{m}}{2}$	$75\,\mathrm{kN}$	

Section and material

IPE 300, values from the profile table.

Symbol	Expression	Value	Note
h		$300\,\mathrm{mm}$	
b		$150\,\mathrm{mm}$	
t_w		$7.1\,\mathrm{mm}$	
t_f		$10.7\,\mathrm{mm}$	
r		$15\,\mathrm{mm}$	
A		$5381\,\mathrm{mm^2}$	
A_v		$2568\,\mathrm{mm^2}$	shear area, z-z
I_y		$8356\cdot10^4\,\mathrm{mm^4}$	
$W_{el,y}$		$557.1\cdot10^3\,\mathrm{mm^3}$	
$W_{pl,y}$		$628.4\cdot10^3\,\mathrm{mm^3}$	
f_y		$355\,\mathrm{MPa}$	
E		$210\,\mathrm{GPa}$	
γ_{M0}		1.0	
γ_{M1}		1.0	

Classification

Outstand flange in compression, Table 5.2.

Symbol	Expression	Value	Note
ε	$\sqrt{\frac{235\,\mathrm{MPa}}{f_y}} = \sqrt{\frac{235\,\mathrm{MPa}}{355\,\mathrm{MPa}}}$	0.8136	
c_f	$\frac{b - t_w - 2r}{2} = \frac{150\,\mathrm{mm} - 7.1\,\mathrm{mm} - 2 \cdot 15\,\mathrm{mm}}{2}$	56.45 mm	
$\frac{c_f}{t_f}$	$\frac{56.45\,\mathrm{mm}}{10.7\,\mathrm{mm}}$	5.276	
if	$\frac{c_f}{t_f} = 5.276 \le 9\varepsilon = 7.322$		
cls		1	

The flange is class 1, so the plastic section modulus is used.

Bending resistance, 6.2.5

Symbol	Expression	Value	Note
$M_{c,Rd}$	$\frac{W_{pl,y}\, f_y}{\gamma_{M0}} = \frac{628.4 \cdot 10^3\,\mathrm{mm}^3 \cdot 355\,\mathrm{MPa}}{1.0}$	223.1 kN·m	(6.13)
check	$M_{Ed} = 112.5\,\mathrm{kN{\cdot}m} \le M_{c,Rd} = 223.1\,\mathrm{kN{\cdot}m}$	**OK · 0.50**	

Shear resistance, 6.2.6

Symbol	Expression	Value	Note
$V_{pl,Rd}$	$\frac{A_v\, f_y}{\sqrt{3}\,\gamma_{M0}} = \frac{2568\,\mathrm{mm}^2 \cdot 355\,\mathrm{MPa}}{\sqrt{3} \cdot 1.0}$	526.3 kN	(6.18)
check	$V_{Ed} = 75\,\mathrm{kN} \le V_{pl,Rd} = 526.3\,\mathrm{kN}$	**OK · 0.14**	
check	$V_{Ed} = 75\,\mathrm{kN} \le 0.5\,V_{pl,Rd} = 263.2\,\mathrm{kN}$	**OK · 0.28**	no M-V interaction, 6.2.8(2)

Deflection

Limit L/250 for the characteristic combination.

Symbol	Expression	Value	Note
δ	$\frac{5\,q_{ser}\,L^4}{384\,E\,I_y} = \frac{5 \cdot 18\,\mathrm{kN/m} \cdot (6.0\,\mathrm{m})^4}{384 \cdot 210\,\mathrm{GPa} \cdot 8356 \cdot 10^4\,\mathrm{mm}^4}$	17.31 mm	
δ_{lim}	$\frac{L}{250} = \frac{6.0\,\mathrm{m}}{250}$	24 mm	
check	$\delta = 17.31\,\mathrm{mm} \le \delta_{lim} = 24\,\mathrm{mm}$	**OK · 0.72**	

Summary

Bending utilization 0.5043, shear 0.1425, deflection 0.7213. IPE 300 S355 is adequate.